

Curve Fitting

Curve Fitting:- It is the method of finding a relation for a given data such an equation is called best Fitting equation.

Curve Fitting by least square method:-

Fitting of a straight line $y = ax + b$.

The equation is of the form $y = ax + b$.

$$y = ax + b \rightarrow (1)$$

$$\sum y = a \sum x + \sum b \rightarrow (2)$$

$$\sum y = a \sum x + nb \rightarrow (2)$$

xy by x

$$\sum xy = a \sum x^2 + b \sum x \rightarrow (3)$$

Here the eqⁿ (2) & (3) are called as normal equations for fitting a straight line $y = ax + b$ in the least square method. By solving normal equations, we obtain the values of a & b.

Alternate form of straight line

$$y = a + bx$$

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

Fitting of a second degree parabola $y = ax^2 + bx + c$

The equation is of the form $y = ax^2 + bx + c \rightarrow (1)$

The normal equations are

$$\Sigma y = a \Sigma x^2 + b \Sigma x + nc \rightarrow (2)$$

$$\Sigma xy = a \Sigma x^3 + b \Sigma x^2 + c \Sigma x \rightarrow (3)$$

$$\Sigma x^2 y = a \Sigma x^4 + b \Sigma x^3 + c \Sigma x^2 \rightarrow (4)$$

Here (2), (3) & (4) are called as normal equations for fitting a 2nd degree parabola $y = ax^2 + bx + c$ by solving these three equations, we can obtain the values of a, b & c .

Alternative form of 2nd degree parabola.

$$y = a + bx + cx^2$$

$$\Sigma y = \Sigma a + b \Sigma x + c \Sigma x^2 \rightarrow (2)$$

$$\Sigma y = na + b \Sigma x + c \Sigma x^2 \rightarrow (2)$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3 \rightarrow (3)$$

$$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4 \rightarrow (4)$$

(2), (3) & (4) are normal equations by ~~solving~~ ^{fitting} these three degree parabola $y = a + bx + cx^2$ by solving these equations we obtain values of a, b & c .

Problems:

1) Fit a straight line $y = ax + b$ for the following data.

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

$$y = ax + b \rightarrow (1)$$

$$\Sigma y = a \Sigma x + nb \rightarrow (2) \quad n = 8$$

$$\Sigma xy = a \Sigma x^2 + b \Sigma x \rightarrow (3)$$

x	y	xy	x^2
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
56	40	364	524

$$(2) \Rightarrow 40 = 56a + 8b$$

$$364 = 524a + 56b.$$

$$\therefore a = 0.636 \quad b = 0.545.$$

$$\therefore y = 0.636x + 0.545$$

This is the best fitting equation of a straight line for given data.

2) Find the equation of straight line for the following data & hence estimate the values of the dependent variable corresponding to the value of 30 of the dependent variable.

x	5	10	15	20	25
y	16	19	23	26	30

consider, $y = a + bx \rightarrow (1)$

$$\sum y = na + b \sum x \quad (n=5) \rightarrow (2)$$

$$\sum xy = a \sum x + b \sum x^2 \rightarrow (3)$$

x	y	xy	x ²
5	16	80	25
10	19	190	100
15	23	345	225
20	26	520	400
25	30	750	625
<u>$\sum x = 75$</u>	<u>$\sum y = 114$</u>	<u>1885</u>	<u>1375</u>

$$(2) \Rightarrow \begin{aligned} 114 &= 5a + 75b \\ 1885 &= 75a + 1375b \end{aligned}$$

$$a = 0.7 \quad b = 12.3$$

$$y = 12.3 + 0.7x$$

when $x = 30$, we obtain

$$y = 12.3 + (0.7)(30)$$

$$y = 33.3$$

3) Fit a st. line to the following data

Year (x)	1961	1971	1981	1991	2001
Production in tons (y)	8	10	12	10	16

also find the expected production in the year 2006.

Solⁿ: let $X = x - 1981$ and the line of fit will be
 $y = a + bX$

The normal Equations are

$$\sum y = an + b \sum X \quad (n=5)$$

$$\sum xy = a \sum x + b \sum x^2$$

x	y	$X = x - 1981$	XY	X^2
1961	8	-20	-160	400
1971	10	-10	-100	100
1981	12	0	0	0
1991	10	10	100	100
2001	16	20	320	400
$\sum y = 56$		$\sum X = 0$	$\sum XY = 160$	$\sum X^2 = 1000$

The normal Equations becomes

$$56 = 5a + 0$$

$$160 = a(0) + b(1000)$$

$$\therefore a = 11.2, \quad b = 0.16$$

The st. line is

$$y = a + bx$$

$$y = 11.2 + 0.16x$$

$$y = 11.2 + 0.16(x - 1981)$$

when $x = 2006$

$$y = 11.2 + 0.16(2006 - 1981)$$

$$y = 15.2$$

H.W.

4) Find the equation of the best fitting st. line for the following data.

a)	x	1	2	3	4	5
	y	14	13	9	5	2

b)	x	62	64	65	69	70	71	72
	y	65.7	66.8	67.2	69.3	69.8	70.5	70.9

c)	year	1911	1921	1931	1941	1951
	production (in thousand)	8	10	12	10	6

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Fitting of a second degree parabola $y = ax^2 + bx + c$

Problems:

- i) Fit a best fitting parabola $y = ax^2 + bx + c$ for the following data.

x	1	2	3	4	5
y	10	12	13	16	19

& hence estimate at $x = 6$.

The normal equations are

$$\sum y = a \sum x^2 + b \sum x + cn \quad (n = 5)$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

x	y	x^2	x^3	x^4	xy	x^2y
1	10	1	1	1	10	10
2	12	4	8	16	24	48
3	13	9	27	81	39	117
4	16	16	64	256	64	256
5	19	25	125	625	95	475
$\sum x = 15$	70	55	225	979	232	906

The normal Equations becomes

$$70 = 55a + 15b + 5c$$

$$232 = 225a + 55b + 15c$$

$$906 = 979a + 225b + 55c$$

$$a = 0.2857 \approx 0.29$$

$$b = 0.4857 \approx 0.49$$

$$c = 9.4$$

The equation of parabola is

$$y = 0.29x^2 + 0.49x + 9.4$$

when $x = 6$; $y = 22.78$

2) Fit a parabola $y = a + bx + cx^2$ for the following data.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	2.3

The normal equations are

$$\sum y = an + b\sum x + c\sum x^2$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$

$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

Here $n = 5$

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	2.3	16	64	256	9.2	36.8
10	8.9	30	100	354	21.1	66.3

The normal equations becomes.

$$8.9 = 5a + 10b + 30c$$

$$21.1 = 10a + 30b + 100c$$

$$66.3 = 30a + 100b + 354c$$

$$a = 1.0771 ; b = 0.4157 ; c = -0.0214$$

The equation $y = 1.0771 + 0.4157x - 0.0214x^2$

H.W.

3) Following are the measurements of air velocity (x) and evaporation co-efficient (y) of burning fuel droplets in an impulse engine.

x	20	60	100	140	180	220
y	0.18	0.37	0.35	0.78	0.56	0.75

Find a best fitting parabola $y = ax^2 + bx + c$ to the above data and hence estimate y when $x = 99$.

4) Fit a best fitting parabola $y = ax^2 + bx + c$ for the following data.

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

Fitting of a curve of the form $y = a \cdot x^b$

$$y = a \cdot x^b$$

Taking log o.b.s

$$\begin{aligned}\log y &= \log (a \cdot x^b) \\ &= \log a + \log x^b\end{aligned}$$

$$\log y = \log a + b \log x$$

This is of the form; $Y = A + BX$

where; $Y = \log y$

$$X = \log x$$

$$A = \log a \Rightarrow a = e^A$$

$$B = b.$$

The normal equations $Y = A + BX$ (str. line)

$$\sum Y = An + B \sum X.$$

$$\sum XY = A \sum X + B \sum X^2.$$

Problems:

1) Fit a best fitting equation in the form $y = ax^b$ for the following data.

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.1	6.8	7.5

Calculate the value y when $x = 3.5$

The equation $y = a \cdot x^b$ will reduce to

$$Y = A + B X$$

The normal equations are

where

$$\begin{aligned}\sum Y &= nA + B \sum X \\ \sum XY &= A \sum X + B \sum X^2\end{aligned}$$

$$Y = \log y$$

$$X = \log x$$

$$A = \log a \Rightarrow a = e^A$$

x	y	$Y = \log y$	$X = \log x$	X^2	XY
1	2.98	1.0912	0	0	0
2	4.26	1.4493	0.6931	0.4803	1.0045
3	5.21	1.6506	1.0986	1.2069	1.8133
4	6.1	1.8083	1.3862	1.9215	2.5066
5	6.8	1.9169	1.6094	2.5901	3.0850
6	7.5	2.0149	1.7917	3.2101	3.6100
		<u>$\sum Y = 9.9312$</u>	<u>$\sum X = 6.5790$</u>	<u>9.4089</u>	<u>12.0194</u>

The normal Equations becomes.

$$9.9312 = 6A + 6.5790B$$

$$12.0194 = 6.5790A + 9.4089B$$

$$A = 1.0908 ; B = 0.5147$$

$$\text{Now, } a = e^A = e^{1.0908} = 2.9766$$

$$\text{The curve is } y = a \cdot x^b = (2.9766) x^{0.5147}$$

$$\text{when } x = 3.5 \quad y = (2.9766) (3.5)^{0.5147}$$

$$\boxed{y = 5.6722}$$

2) Fit a least square geometric curve $y = ax^b$ for the following data.

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

consider $y = ax^b$

$$\log y = \log a + b \log x$$

$$Y = A + BX$$

$$\sum Y = nA + B \sum X$$

$$\sum XY = A \sum X + B \sum X^2$$

$$Y = \log y$$

$$A = \log a$$

$$X = \log x$$

x	y	$X = \log x$	$Y = \log y$	XY	X^2
1	0.5	0	-0.6931	0	0
2	2	0.6931	0.6931	0.4804	0.4804
3	4.5	1.0986	1.5041	1.6524	1.2069
4	8	1.3863	2.0794	2.8827	1.9218
5	12.5	1.6094	2.5257	4.0649	2.5902
		<u>4.7874</u>	<u>6.1092</u>	<u>9.0804</u>	<u>6.1993</u>

$$6.1092 = 5A + 4.7874B$$

$$9.0804 = 4.7874A + 6.1993B$$

$$A = -0.69315 \quad B = 2$$

$$A = \log_e a \Rightarrow a = e^A = e^{-0.69315} = 0.5$$

$y = 0.5 x^2$ is the required curve.

3) An experiment on life time t of cutting tool at different cutting speed V (units) is given below.

Speed (V)	350	400	500	600
Life (t)	61	26	7	2.6

Fit a relation of the form $V = at^b$

Consider $V = at^b$

$$\log V = \log a + b \log t$$

$$V = A + bT$$

$$V = \log v$$

$$A = \log a$$

$$T = \log t$$

$$\sum V = nA + b \sum T$$

$$\sum VT = A \sum T + b \sum T^2$$

V	t	$V = \log v$	$T = \log t$	VT	T^2
350	61	5.8579	4.1109	24.0812	16.8995
400	26	5.9915	3.2581	19.5209	10.6152
500	7	6.2146	1.9459	12.093	3.7865
600	2.6	6.3969	0.9555	6.1122	0.913
		24.4609	10.2704	61.8073	32.2124

$$4A + 10.2704 B = 24.4609$$

$$10.2704 A + 32.2124 B = 61.8073$$

$$A = 6.5539 \quad B = -0.1709$$

$$A = \log a \Rightarrow a = e^A = e^{6.5539} = 701.9766$$

$$y = (701.9766) t^{-0.1709}$$

H.W.

4) Fit a curve of the form $y = ax^b$ for the following data

x	5	6	7	8	9	10
y	133	55	23	7	2	2

correlation and correlation coefficient :-

- * co-variation of two independent magnitudes is known as correlation.
- * If two variables x & y are related in such way that $\uparrow$ (or) $\downarrow$ in one of them corresponds to $\uparrow$ (or) $\downarrow$ in the other,
- * We say that the variables are +vely co-related.
- * Also if $\uparrow$ (or) $\downarrow$ in one of them corresponds to $\downarrow$ (or) $\uparrow$ in the other. the variables are said to be -vely ~~cor~~ correlated.

The numerical measure of correlation b/w two variables ' x ' & ' y ' is known as Pearson's co-efficient of correlation or Karl's pearson co-efficient of correlation usually denoted by ' r ' and is defined by

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \cdot \sqrt{\sum y^2}}$$

where $x = x - \bar{x}$

$$y = y - \bar{y}$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{y} = \frac{\sum y}{n}$$

NOTE:

- 1) The co-efficient of correlation numerically does not exceed unity i.e., $-1 \leq r \leq 1$
- 2) If $r = \pm 1$, we say that x & y are perfectly correlated
- 3) If $r = 0$, then we say that ' x ' & ' y ' are non-correlated.

Regression :-

Regression is an estimation of one independent variable in terms of the other. If ' x ' & ' y ' are correlated, the best fitting st. line in the least square sense gives reasonably a good relation b/w x & y .

The best fitting st. line of the form $y = ax + b$ [x being the independent variable] is called the regression line of y on x .
 $x = ay + b$ [y being the independent variable] is called the regression line x on y .

Formula to find the lines of Regression:-

Equation of line of regression of 'y' on 'x'
 $(y - \bar{y}) = b_{yx} (x - \bar{x})$ where $b_{yx} = r \cdot \frac{\sigma_x}{\sigma_y}$

Equation of line of regression of 'x' on 'y'
 $(x - \bar{x}) = b_{xy} (y - \bar{y})$ where $b_{xy} = r \cdot \frac{\sigma_y}{\sigma_x}$

$$\sigma_x = \sqrt{\frac{\sum X^2}{n}}$$

$$\sigma_y = \sqrt{\frac{\sum Y^2}{n}}$$

NOTE:-

To find co-efficient of correlation 'r' by using lines of regression is

$$r = \pm \sqrt{b_{xy} \times b_{yx}}$$

Problems:

1) Compute the co-efficient of correlation and the equation of the lines of regression for the data.

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

Here $n = 7$

$$\bar{x} = \frac{\sum x}{n} = \frac{1+2+3+4+5+6+7}{7} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{\sum y}{n} = \frac{9+8+10+12+11+13+14}{7} = \frac{77}{7} = 11$$

x	y	$x = x - \bar{x} = x - 4$	$y = y - \bar{y} = y - 11$	xy	x^2	y^2
1	9	-3	-2	6	9	4
2	8	-2	-3	6	4	9
3	10	-1	-1	1	1	1
4	12	0	1	0	0	1
5	11	1	0	0	1	0
6	13	2	2	4	4	4
7	14	3	3	9	9	9
		$\sum x = 0$	$\sum y = 0$	$\sum xy = 26$	28	28

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}}$$

$$r = \frac{26}{\sqrt{28} \cdot \sqrt{28}} = \frac{26}{28} = 0.9285$$

$$r = 0.9285 \approx 0.93$$

$$\sigma_x = \sqrt{\frac{\sum x^2}{n}} = \sqrt{\frac{28}{7}} = \sqrt{4} = 2$$

$$\sigma_y = \sqrt{\frac{\sum y^2}{n}} = \sqrt{\frac{28}{7}} = \sqrt{4} = 2$$

Equation of line of regression of y on x

$$y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 11 = (0.93) \left(\frac{3}{8} \right) (x - 4)$$

$$y - 11 = 0.93x - 3.70$$

$$y = 0.93x - 3.70 + 11$$

$$\boxed{y = 0.93x + 7.30}$$

Equation of line of regression of x on y

$$x - \bar{x} = r \cdot \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 4 = (0.93) \left(\frac{3}{2} \right) (y - 11)$$

$$x - 4 = 0.93y - 10.23$$

$$x = 0.93y - 10.23 + 4$$

$$\boxed{x = 0.93y - 6.23}$$

2) The following table gives the heights of fathers (x) & sons (y)

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	70	69	71

Calculate the co-efficient of correlation and lines of regression.

Here $n=8$

$$\bar{x} = \frac{\sum x}{n} = \frac{65+66+67+67+68+69+70+72}{8} = 68$$

$$\bar{y} = \frac{\sum y}{n} = \frac{67+68+65+68+70+70+69+71}{8} = 69$$

x	y	$x - \bar{x} = x - 68$	$y - \bar{y} = y - 69$	xy	x^2	y^2
65	67	-3	-2	6	9	4
66	68	-2	-1	2	4	1
67	65	-1	-4	4	1	16
67	68	-1	-1	1	1	1
68	70	0	3	0	0	9
69	70	1	3	3	1	9
70	69	2	0	0	4	0
72	71	4	2	8	16	4
				$\sum xy = 24$	$\sum x^2 = 36$	$\sum y^2 = 44$

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}}$$

$$r = \frac{24}{\sqrt{36} \cdot \sqrt{44}} = \frac{24}{39.7994} = 0.6030$$

$$\sqrt{0.171644}$$

$$r = 0.6030$$

$$\sigma_x = \sqrt{\frac{\sum x^2}{n}} = \sqrt{\frac{36}{8}} = 2.1213$$

$$\sigma_y = \sqrt{\frac{\sum y^2}{n}} = \sqrt{\frac{44}{8}} = 2.3452$$

Equation of line of regression of y on x

$$y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 69 = (0.6030) \left(\frac{2.34}{2.121} \right) (x - 68)$$

$$y - 69 = (0.6652) (x - 68)$$

$$= 0.6652x - 45.2336$$

$$y = 0.6652x - 45.2336 + 69$$

$$\boxed{y = 0.6652x + 23.7664}$$

Equation of line of regression of x on y

$$x - \bar{x} = r \cdot \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 68 = (0.6030) \left(\frac{2.121}{2.34} \right) (y - 69)$$

$$x - 68 = (0.5465) (y - 69)$$

$$x - 68 = 0.5465y - 37.713$$

$$x = 0.5465y - 37.713 + 68$$

$$\boxed{x = 0.5465y + 30.286}$$

3) obtain the lines of regression & hence find the co-efficient of correlation for the data.

x	1	3	4	5	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

Solution

$$n = 10$$

$$\bar{x} = \frac{\sum x}{n} = \frac{1+3+4+2+5+8+9+10+13+15}{10} = \frac{70}{10} = 7$$

$$\bar{y} = \frac{\sum y}{n} = \frac{8+6+10+8+12+16+16+10+32+32}{10} = \frac{150}{10} = 15$$

x	y	$x - \bar{x} = x - 7$	$y - \bar{y} = y - 15$	xy	x^2	y^2
1	8	-6	-7	42	36	49
3	6	-4	-9	36	16	81
4	10	-3	-5	15	9	25
2	8	-5	-7	35	25	49
5	12	-2	-3	6	4	9
8	16	1	1	1	1	1
9	16	2	1	2	4	1
10	10	3	-5	-15	9	25
13	32	6	17	102	36	289
15	32	8	17	136	64	289
				360	204	818

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}} = \frac{360}{\sqrt{204} \sqrt{818}} = \frac{360}{(14.28)(28.6006)} = \frac{360}{408.489}$$

$$\boxed{r = 0.881}$$

$$\sigma_x = \sqrt{\frac{\sum x^2}{n}} = \sqrt{\frac{204}{10}} = 4.516$$

$$\sigma_y = \sqrt{\frac{\sum y^2}{n}} = \sqrt{\frac{818}{10}} = 9.044$$

Equation of line of regression of y on x

$$y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 15 = (0.881) \left(\frac{9.044}{4.516} \right) (x - 7)$$

$$= (1.7644)(x - 7)$$

$$y - 15 = 1.7644x - 12.350$$

$$y = 1.7644x - 12.350 + 15$$

$$\boxed{y = 1.7644x + 2.649}$$

Equation of line of regression of x on y

$$x - \bar{x} = r \cdot \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 7 = (0.881) \left(\frac{4.516}{9.044} \right) (y - 15)$$

$$= (0.4399)(y - 15)$$

$$x - 7 = 0.4399y - 6.5987$$

$$x = 0.4399y - 6.5987 + 7$$

$$\boxed{x = 0.4399y + 0.4012}$$

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4) The following data gives the age of husband (x), the age of wife (y) in years. Form the 2 regression lines and calculate the age of husband and corresponding 16 years age of wife.

x	36	23	27	28	28	29	30	31	33	35
y	29	18	20	22	27	21	29	27	29	28

Soln Regression lines are.

$$y + \bar{y} =$$

$$n = 10.$$

$$\bar{x} = \frac{\sum x}{n} = \frac{36 + 23 + 27 + 28 + 28 + 29 + 30 + 31 + 33 + 35}{10} = 30$$

$$\bar{y} = \frac{\sum y}{n} = \frac{29 + 18 + 20 + 22 + 27 + 21 + 29 + 27 + 29 + 28}{10} = 25$$

x	y	$X = x - \bar{x} = x - 30$	$Y = y - \bar{y} = y - 25$	XY	X^2	Y^2
36	29	6	4	24	36	16
23	18	-7	-7	49	49	49
27	20	-3	-5	15	9	25
28	22	-2	-3	6	4	9
28	27	-2	2	-4	4	4
29	21	-1	-4	4	1	16
30	29	0	4	0	0	16
31	27	1	2	2	1	4
33	29	3	4	12	9	16
35	28	5	3	15	25	9
				<u>123</u>	<u>138</u>	<u>164</u>

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}} = \frac{103}{(11.747)(12.806)} = 0.8176 \approx 0.82$$

$$\bar{x} = \sqrt{\frac{\sum X^2}{n}} = \sqrt{\frac{138}{10}} = 3.714$$

$$\bar{y} = \sqrt{\frac{\sum Y^2}{n}} = \sqrt{\frac{164}{10}} = 4.0496$$

Equation of line of regression of y on x

$$y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 25 = 0.82 \left(\frac{4.0496}{3.714} \right) (x - 30)$$

$$y - 25 = (0.894)(x - 30)$$

$$y - 25 = 0.894x - 26.82$$

$$y = 0.894x - 26.82 + 25$$

$$y = 0.894x - 1.82$$

$$\boxed{y = 0.894x - 1.82}$$

Equation of line of regression of x on y

$$x - \bar{x} = r \cdot \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 30 = 0.82 \left(\frac{3.714}{4.0496} \right) (y - 25)$$

$$= (0.7520)(y - 25)$$

$$x - 30 = 0.7520y - 18.8$$

$$x = 0.7520y - 18.8 + 30$$

$$\boxed{x = 0.7520y + 11.2}$$

when $y = 16$, $x = ?$

$$x = 0.7520(16) + 11.2$$

$$x = 12.032 + 11.2$$

$$x = 23.232 \approx 23$$

$$\boxed{x = 23}$$

1) With usual notations, compute $\bar{x}$, $\bar{y}$ & r from the following lines of regression.

$$2x + 3y + 1 = 0 \quad \& \quad x + 6y - 4 = 0$$

Solⁿ: We know the lines of regression passes through $\bar{x}$ & $\bar{y}$.

$$\Rightarrow 2\bar{x} + 3\bar{y} + 1 = 0 \quad \& \quad \bar{x} + 6\bar{y} - 4 = 0$$

$$2\bar{x} + 3\bar{y} = -1 \quad \& \quad \bar{x} + 6\bar{y} = 4$$

(At last)

$$\therefore \bar{x} = -2, \quad \bar{y} = 1$$

We assume that $2x + 3y + 1 = 0$ is line of regression y on x

$$3y = -2x - 1$$

$$y = -\frac{2}{3}x - \frac{1}{3}$$

$$b_{yx} = -\frac{2}{3}$$

We assume that $x + 6y - 4 = 0$ is line of regression x on y

$$x = -6y + 4$$

$$b_{xy} = -6$$

$$\therefore r = \pm \sqrt{-\frac{2}{3} \times -6} = 2 > 1$$

$\therefore$ The assumption is wrong.

So, $2x + 3y + 1 = 0$ is a line of regression x on y
 $x + 6y - 4 = 0$ is a line of regression y on x

$$2x + 3y + 1 = 0$$

$$2x = -\frac{3}{2}y - \frac{1}{2}$$

$$b_{xy} = -\frac{3}{2}$$

$$x + 6y - 4 = 0$$

$$6y = -x + 4$$

$$y = -\frac{1}{6}x + \frac{4}{6}$$

$$b_{yx} = -\frac{1}{6}$$

$$\therefore r = \pm \sqrt{b_{xy} \times b_{yx}}$$

$$r = \pm \sqrt{\left(-\frac{3}{5}\right)\left(-\frac{1}{2}\right)}$$

$$r = \pm \sqrt{\frac{1}{4}}$$

$$r = \pm \frac{1}{2}$$

$$r = \pm 0.5 < 1$$

$$\boxed{r = 0.5}$$

Q) With usual notation, compute $\bar{x}$, $\bar{y}$ & r from the following lines of regression.

$$y = 0.516x + 33.73 \quad \& \quad x - 30.50 = 0.512y + 30.52$$

Soln: W.K.T the lines of regression, passes through $\bar{x}$ & $\bar{y}$

$$\bar{y} = 0.516\bar{x} + 33.73$$

↓
y on x

$$b_{yx} = 0.516$$

$$\bar{x} = 0.512\bar{y} + 65.04$$

↓
x on y

$$b_{xy} = 0.512$$

$$r = \pm \sqrt{b_{yx} \cdot b_{xy}}$$

$$r = \pm \sqrt{(0.516)(0.512)}$$

$$r = \pm 0.5139$$

$$\therefore \boxed{r = 0.5139}$$

The regression lines can be written as

$$\begin{cases} 0.516\bar{x} - \bar{y} = -33.73 \\ \bar{x} - 0.512\bar{y} = 65.04 \end{cases}$$

$$\bar{x} = 111.86, \quad \bar{y} = 91.45$$

3) Given equation of regression lines $x = 19.13 - 0.81y$
 $y = 11.64 - 0.5x$ compute $\bar{x}$, $\bar{y}$ & r .

W.K.T Regression lines passes through $\bar{x}$ & $\bar{y}$

$$\bar{x} = 19.13 - 0.81\bar{y}$$

$$\bar{x} = -0.81\bar{y} + 19.13$$

↓

x on y

$$b_{xy} = -0.81$$

$$\bar{y} = 11.64 - 0.5\bar{x}$$

$$\bar{y} = -0.5\bar{x} + 11.64$$

↓

y on x

$$b_{yx} = -0.5$$

$$r = \pm \sqrt{b_{xy} \cdot b_{yx}}$$

$$r = \pm \sqrt{(-0.81)(-0.5)}$$

$$r = \pm 0.66$$

$$\boxed{r = 0.66}$$

The regression lines can be written as.

$$\begin{cases} \bar{x} + 0.81\bar{y} = 19.13 \\ 0.5\bar{x} + \bar{y} = 11.64 \end{cases}$$

$$\bar{x} = 15.94 \quad \bar{y} = 3.67$$

3) Show that if θ is the angle b/w the lines of regression then $\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right)$

Proof: W.K.T if θ is an acute, the angle b/w the lines $y = m_1 x + c_1$ and $y = m_2 x + c_2$ is given by.

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2} \longrightarrow (1)$$

We have the lines of regression

$$y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \longrightarrow (2)$$

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \longrightarrow (3)$$

From (3) $y - \bar{y} = \frac{1}{r} \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \longrightarrow (4)$

Slopes of (2) & (4) are respectively.

$$m_1 = r \cdot \frac{\sigma_y}{\sigma_x} \quad , \quad m_2 = \frac{1}{r} \frac{\sigma_y}{\sigma_x}$$

$$\tan \theta = \frac{\frac{1}{r} \frac{\sigma_y}{\sigma_x} - r \frac{\sigma_y}{\sigma_x}}{1 + r \frac{\sigma_y}{\sigma_x} \cdot \frac{1}{r} \frac{\sigma_y}{\sigma_x}}$$

$$= \frac{\frac{\sigma_y}{\sigma_x} \left(\frac{1}{r} - r \right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}}$$

$$= \frac{\frac{\sigma_y}{\sigma_x}}{\frac{\sigma_x^2 + \sigma_y^2}{\sigma_x^2}} \left(\frac{1-r^2}{r} \right) = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right)$$

4) In a bivariate distribution, it is found that $\sigma_x = \sigma_y$ and the acute angle b/w the lines of regression is $\tan^{-1}(3)$. Find the correlation coefficient.

Soln: Given $\sigma_x = \sigma_y$
 $\theta = \tan^{-1}(3)$
 $\tan \theta = 3$.

We have,

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1-r^2}{r} \right)$$

$$3 = \frac{\sigma_x^2}{2\sigma_x^2} \left(\frac{1-r^2}{r} \right)$$

$$3 = \frac{1}{2} \left(\frac{1-r^2}{r} \right)$$

$$6r = 1-r^2$$

$$r^2 + 6r - 1 = 0$$

$$r = 0.1622 \quad (\text{or}) \quad r = -6.1622 < -1$$

[This is not possible]

$$\therefore \boxed{r = 0.1622}$$

Rank correlation (without repetitions)

(or)

Spearman's rank correlation :-

The Spearman's rank correlation is defined as the Pearson correlation coefficient between the rank variables and it will be denoted by the symbol ρ (row)

$$\rho = 1 - \frac{6 \sum d^2}{n^3 - n}$$

Suppose x & y are the marks scored in the two subjects, let R_x & R_y are the ranks, $d = R_x - R_y$ be the differences of the ranks then the Spearman's rank correlation

$$\rho = 1 - \frac{6 \sum d^2}{n^3 - n}$$

Problems:

Type-1: When the ranks are given

i) The ranking of 10 students in 2 subjects, field theory (A) & network analysis (B) are given below.

Roll no. of the students	1	2	3	4	5	6	7	8	9	10
A	3	5	8	4	7	10	10	1	6	9
B	6	4	9	8	1	2	3	10	5	7

calculate the rank correlation coefficient

Solution

R_A	R_B	$d = R_A - R_B$	d^2
3	6	-3	9
5	4	1	1
8	9	-1	1
4	8	-4	16
7	1	6	36
10	2	8	64
2	3	-1	1
1	10	-9	81
6	5	1	1
9	7	2	4

$$\sum d^2 = 214.$$

The Spearman's rank correlation coefficient

$$\rho = 1 - \frac{6 \sum d^2}{n^3 - n}$$

Here $n = 10$;

$$\rho = 1 - \frac{6(214)}{10^3 - 10}$$

$$= 1 - \frac{1284}{990}$$

$$= 1 - 1.2967$$

$$\boxed{\rho = -0.2967}$$

Type 2:- When ranks are not given

The scores for 10 students in English and mathematics are as follows.

English 56 75 45 71 62 64 58 80 76 61

Mathematics 66 70 40 60 65 56 59 77 67 63

calculate rank correlation co-efficient.

Solution:-

Eng (x)	Maths (y)	R_x	R_y	$d = R_x - R_y$	d^2
56	66	9	4	5	25
75	70	3	2	1	1
45	40	10	10	0	0
71	60	4	7	-3	9
62	65	6	5	1	1
64	56	5	9	-4	16
58	59	8	8	0	0
80	77	1 (1st Rank)	0	0	0
76	67	2 (2nd Rank)	3	-1	1
61	63	7	6	1	1

Spearman's rank correlation coefficient

$$r = 1 - \frac{6 \sum d^2}{n^3 - n}$$

Here $n=10$; $r = 1 - \frac{6(54)}{10^3 - 10}$

$$= 1 - \frac{324}{990} = 1 - 0.3273$$

$$\boxed{r = 0.6727}$$

2) Ten competitors in a beauty contest are ranked by two Judges A & B in the following table.

ID No. of competitors	1	2	3	4	5	6	7	8	9	10
Judge A	1	6	5	3	10	2	4	9	7	8
Judge B	6	4	9	8	1	2	3	10	5	7

Calculate the rank correlation coefficient.

Solution

R_A	R_B	$d = R_A - R_B$	d^2
1	6	-5	25
6	4	2	4
5	9	-4	16
3	8	-5	25
10	1	9	81
2	2	0	0
4	3	1	1
9	10	-1	1
7	5	2	4
8	7	1	1
			$\Sigma d^2 = 158$

The spearman's rank correlation coefficient is

$$r = 1 - \frac{6 \Sigma d^2}{n^3 - n}$$

$$\text{Here } n = 10; \quad r = 1 - \frac{6(158)}{10^3 - 10} = 1 - \frac{948}{990}$$

$$r = 1 - 0.9576$$

$$\boxed{r = 0.0424}$$

3) A random sample of recent repair jobs was selected and estimated cost and actual cost were recorded.

Estimated cost (x)	300	450	800	250	500	975	475	400
Actual cost (y)	273	486	734	297	631	872	396	457

compute rank correction co-efficient.

Solution:

x	y	R _x	R _y	d = R _x - R _y	d ²
300	273	7	8	-1	1
450	486	5	4	1	1
800	734	2	2	0	0
250	297	8	7	1	1
500	631	3	3	0	0
975	872	1	1	0	0
475	396	4	6	-2	4
400	457	6	5	1	1

Spearman's rank correlation coefficient

$$\rho = 1 - \frac{6 \sum d^2}{n^3 - n}$$

Here $n=8$, $\rho = 1 - \frac{6(8)}{8^3 - 8}$

$$\rho = 1 - \frac{48}{504}$$

$$= 1 - 0.095$$

$$\boxed{\rho = 0.9047}$$

4) 10 students got the following percentage of marks in 2 subjects x & y. Find their rank correlation co-efficient

Marks in (x) 78 36 98 25 75 82 90 62 65 39

Marks in (y) 84 50 91 60 68 62 86 58 53 47

Marks in (x)	Marks in (y)	Rank in (x)	Rank in (y)	$d = R_x - R_y$	d^2
78	84	4	3	1	1
36	50	9	9	0	0
98	91	1	1	0	0
25	60	10	6	4	16
75	68	5	4	1	1
82	62	3	5	-2	4
90	86	2	2	0	0
62	58	7	7	0	0
65	53	6	8	-2	4
39	47	8	10	-2	4
					$\Sigma d^2 = 30$

Spearman's rank correlation coefficient

$$\rho = 1 - \frac{6 \Sigma d^2}{n^3 - n}$$

Here $n = 10$;

$$\rho = 1 - \frac{6(30)}{10^3 - 10}$$

$$\rho = 1 - \frac{180}{990}$$

$$\boxed{\rho = 0.82}$$

5) 10 competitors in music contest are ranked by 3 judges A, B, C in the following order. Use the rank correlation coefficient to decide which pair of judges have nearest approach to common taste of music.

A	1	6	5	10	3	2	4	9	7
B	3	5	8	4	7	10	2	1	6
C	6	4	9	8	1	2	3	10	5

Solution: We shall find ρ_{AB} , ρ_{BC} , ρ_{CA} .

A	B	C	d_{AB}	d_{BC}	d_{CA}	d_{AB}^2	d_{BC}^2	d_{CA}^2
1	3	6	-2	-3	5	4	9	25
6	5	4	1	1	-2	1	1	4
5	8	9	-3	-1	+4	9	1	16
10	4	8	6	-4	-2	36	16	4
3	7	1	-4	6	-2	16	36	4
2	10	2	-8	8	0	64	64	0
4	2	3	2	-1	-1	4	1	1
9	1	10	8	-9	1	64	81	1
7	6	5	1	1	-2	1	1	4
8	9	7	-1	-2	-1	1	4	1
						$\sum d_{AB}^2 = 200$	$\sum d_{BC}^2 = 214$	$\sum d_{CA}^2 = 60$

Here $n = 10$

$\rho_{AB} = 1 - \frac{6(200)}{10^3 - 10} = -0.21$	It may be observed that ρ_{AB} & ρ_{BC} are -ve. which means that their tastes (A & B, B & C) are opposite. But ρ_{CA} is +ve and is nearer to 1 (perfect correlation)
$\rho_{BC} = 1 - \frac{6(214)}{10^3 - 10} = -0.297$	
$\rho_{CA} = 1 - \frac{6(60)}{10^3 - 10} = 0.636$	

HW

- 1) The scores for 9 students in physics (x) & mathematics (y) are as follows.

x	35	23	47	17	10	43	9	6	28
y	30	33	45	23	8	49	10	4	31

compute the Ranks & Rank correlation

- 2) The participants in a contest are ranked by 2 judges as follows.

x	1	6	5	3	10	2	4	9	7	8
y	6	4	9	8	1	2	3	10	5	7

$$\boxed{\rho = 0.042}$$